

Exact Two-Dimensional $SU(2)$ Yang–Mills in Lean: Weyl Integration, Heat-Kernel Convolution, Migdal Invariance, and the Exact Simple-Loop Area Law

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Abstract

We give a kernel-checked Lean 4/Mathlib formalization of the exact heat-kernel solution of two-dimensional $SU(2)$ Yang–Mills theory along the chain from Haar measure to a concrete Migdal subdivision move. The development identifies the transported Haar probability on $SU(2) \simeq S^3$ with the canonical spherical probability, proves the required all-order orbital integration law, derives translated character convolution for every pair of irreducible labels, passes from finite character sums to the infinite heat-kernel semigroup by dominated convergence, and integrates an actual edge shared by two faces. We then define finite oriented disk cellulations independently of any reduction tree, using vertices, paired half-edges, cyclic face words, incidence, Euler characteristic, and positive face areas. Every connected dual graph admits a certified elimination schedule, every valid schedule reduces to the heat kernel at total area, and any two choices give the same amplitude. For the first cyclic unreduced lattice, a three-face disk, we additionally construct a triangular gauge-fixing equivalence that preserves the original triple product Haar measure and prove equality with the reduced amplitude. The same measure-theoretic construction is then proved uniformly for one vertex of arbitrary finite valence. We further construct a rooted spanning-tree order for every finite connected primal graph and a single global root-plus-edge-increments equivalence on all cellulation vertex variables, with exact product-Haar preservation. The schedule is hidden from the public loop object, which supplies a non-trivial exact-area-law package. Combined with all-label coefficient extraction, this yields the exact normalized simple-loop identity

$$\mathbb{E}[W_n(C)] = \exp\left[-\frac{n(n+2)}{4} \text{Area}(C)\right].$$

All displayed endpoints compile without `sorry`, `admit`, or project-local axioms. The mathematical identities are classical; the claimed contribution is their concrete end-to-end machine-checked composition, not a new analytic solution of two-dimensional Yang–Mills theory.

1 Statement and claim boundary

Let dg be normalized Haar probability on $SU(2)$, let χ_n be the character of the irreducible representation of dimension $d_n = n + 1$, and put

$$c_n = \frac{n(n+2)}{4}, \quad K_t(g) = \sum_{n \geq 0} d_n e^{-tc_n} \chi_n(g), \quad t > 0.$$

With convolution

$$(f * h)(g) = \int_{SU(2)} f(x) h(x^{-1}g) dx,$$

the formalized chain is

$$\begin{aligned} \text{Haar on } S^3 &\implies \text{all-order orbital law} \implies \chi_n * \chi_m \\ &\implies K_s * K_t = K_{s+t} \implies \text{Migdal} \implies \text{simple-loop coefficient}. \end{aligned}$$

This result is distinct from the programme’s earlier strong-coupling area-law theorem. The earlier theorem is a volume-uniform *inequality* for finite lattices and $SU(N_c)$ in an explicit convergence window, obtained from a Kotecký–Preiss cluster expansion. The present theorem is an *exact identity* in the two-dimensional $SU(2)$ heat-kernel model, obtained by Haar integration and subdivision. The former is cited context; it is not used to discharge any exact two-dimensional statement here.

No claim is made here about four-dimensional Yang–Mills, a continuum construction in four dimensions, or a new proof of the classical exact solution. A priority claim about formalization should remain conditional on a systematic literature and repository search.

2 From spherical Haar measure to the orbital law

Write an element of $SU(2)$ in its standard form

$$g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}, \quad |a|^2 + |b|^2 = 1.$$

Earlier modules construct a homeomorphism $SU(2) \simeq S^3$, realize left multiplication as an ambient real-linear isometry, prove preservation of the canonical `Measure.toSphere` measure, and use uniqueness of normalized Haar measure to obtain literal equality of the two probability measures.

The new all-order step studies $u = |a|^2$. In dimension four, the squared mass in one complex rail is uniform on $[0, 1]$. Rather than assume this distribution, the formal proof establishes its moment recurrence, proves

$$\int_{SU(2)} u^k dg = \frac{1}{k+1} \quad (k \geq 0),$$

and then identifies measures on the compact interval using polynomial density. The resulting literal endpoint is:

```
theorem su2FirstRailMassMeasure_eq_uniform :
  su2FirstRailMassMeasure = su2UnitIntervalMeasure
```

For $x, y \in [-1, 1]$, choose representatives with half-traces x, y and average their product over conjugation. The preceding uniform law reduces the orbital integral to a one-variable Chebyshev integral. For the Chebyshev polynomial U_m of the second kind, the formalized formula is

$$\int_{SU(2)} U_m\left(\frac{1}{2} \text{tr}(kak^{-1}b)\right) dk = \frac{U_m(\frac{1}{2} \text{tr } a) U_m(\frac{1}{2} \text{tr } b)}{m+1}.$$

The proof includes the endpoint cases $x = \pm 1$; no division by a vanishing sine is hidden in the statement.

3 All-order translated character convolution

Characters are implemented concretely as

$$\chi_n(g) = U_n\left(\frac{1}{2} \text{tr } g\right).$$

Conjugacy invariance, left and right Haar invariance, Fubini, and the orbital law give the translated orthogonality relation

$$\int_{SU(2)} \chi_n(x) \chi_m(x^{-1}g) dx = \delta_{nm} \frac{\chi_n(g)}{n+1}. \quad (1)$$

This is stronger than orthogonality at the identity and is exactly the consumer needed by the heat-kernel product.

Theorem 3.1 (Machine-checked character convolution). *For every $n, m \in \mathbb{N}$ and $g \in SU(2)$,*

$$(\chi_n * \chi_m)(g) = \begin{cases} \chi_n(g)/(n+1), & n = m, \\ 0, & n \neq m. \end{cases}$$

The public Lean endpoints are listed below. They are theorems about the actual normalized Haar integral, not fields projected from an abstract character package.

```
integral_su2CharacterReal_mul_translate
su2CharacterChebyshev_convolution
```

4 Infinite heat-kernel semigroup

For a cutoff N , expand both finite kernels, interchange the finite sums and the Haar integral, and apply (1). The Kronecker selector leaves a single sum and the Casimir weights multiply:

$$e^{-sc_n} e^{-tc_n} = e^{-(s+t)c_n}.$$

This proves the finite convolution identity. The infinite theorem then uses the already established uniform Casimir majorant for positive times. Partial kernels converge pointwise and uniformly; the product is bounded by the product of two summable majorants, so dominated convergence passes the Haar integral to the limit.

Theorem 4.1 (Concrete heat semigroup). *For $s, t > 0$ and every $g \in SU(2)$,*

$$\int_{SU(2)} K_s(x) K_t(x^{-1}g) dx = K_{s+t}(g).$$

*Equivalently, as functions, $K_s * K_t = K_{s+t}$.*

The corresponding Lean declarations, including the function-level consumer, are:

```
su2HeatKernelPartial_convolution
su2HeatKernel_convolution
su2HeatKernel_convolution_consumer
```

5 A concrete Migdal move

Consider two adjacent faces of areas s and t . Let a, b encode the external boundary holonomies and integrate their shared oriented edge x . The actual two-face density in the development is

$$D_{s,t}^{a,b}(x) = K_s(ax) K_t(x^{-1}b).$$

Left invariance changes variables $y = ax$ and the heat semigroup gives

$$\int_{SU(2)} D_{s,t}^{a,b}(x) dx = K_{s+t}(ab). \quad (2)$$

Theorem 5.1 (Two-face Migdal subdivision). *Equation (2) holds for all $a, b \in SU(2)$ and all $s, t > 0$. It is proved by a genuine Haar integration over the shared edge and invokes the concrete infinite heat-kernel semigroup.*

This closes the first nontrivial subdivision move: two holonomy variables, two faces, and one integrated shared edge. The endpoints below contain no package hypothesis.

```
su2Migdal_twoFace_merge
su2Migdal_subdivision_invariant
```

6 A cyclic product-Haar gauge fixing

The two-face move alone does not identify a reduced evaluator with the original lattice integral when the face-dual graph has cycles. We therefore formalize the first cyclic test explicitly. Consider a disk split into three bounded faces by three spokes meeting at one interior vertex. Its face dual is K_3 , equivalently the three-cycle. Write $x, y, z \in SU(2)$ for the original spoke-edge variables and $a, b, c \in SU(2)$ for the fixed boundary arcs. The three face holonomies are

$$xay^{-1}, \quad ybz^{-1}, \quad zcx^{-1}.$$

Thus the unreduced density and measure are

$$K_{t_1}(xay^{-1})K_{t_2}(ybz^{-1})K_{t_3}(zcx^{-1}) dx dy dz. \quad (3)$$

Gauge transformations at the interior vertex act by simultaneous left multiplication. Lean constructs the measurable equivalence

$$\Phi(x, y, z) = (x, x^{-1}y, x^{-1}z), \quad \Phi^{-1}(u, v, w) = (u, uv, uw).$$

The skew-product theorem and left invariance of normalized Haar prove the literal measure identity

$$\Phi_*(dx dy dz) = dx dy dz.$$

After substitution, conjugation invariance of the heat kernel removes the first coordinate and gives

$$K_{t_1}(av^{-1})K_{t_2}(vbw^{-1})K_{t_3}(wc).$$

The pure-gauge integral has mass one. Applying the reversed two-face Migdal identity first to w and then to v yields the fully unreduced result

$$\int_{SU(2)^3} K_{t_1}(xay^{-1})K_{t_2}(ybz^{-1})K_{t_3}(zcx^{-1}) dx dy dz = K_{t_1+t_2+t_3}(abc). \quad (4)$$

The proof does not store a Jacobian certificate or assume an abstract gauge package. The map, its inverse, measurability, product-Haar preservation, density factorization, Fubini integrability, and both edge integrations are all kernel checked. The principal endpoints are:

```
su2ThreeSpokeFaceDual_connected
su2ThreeSpokeGaugeFix_measurePreserving
su2ThreeSpoke_density_gaugeFactorization
su2ThreeSpoke_unreducedIntegral_eq_gaugeFixedIntegral
su2ThreeSpoke_unreducedIntegral_eq_heatKernel
```

Equation (4) is a genuine product-Haar theorem for a cyclic dual, but it is intentionally not stated as a theorem for arbitrary finite cellulations. That uniform generalization requires a global spanning tree gauge fixing compatible with the incidence data of the independent disk object.

6.1 Uniform finite-valence theorem

The preceding three-spoke map is not intrinsically three-dimensional. Let I be any finite type and let

$$d\mu_I = \bigotimes_{i \in I} dg_i$$

be the normalized product Haar probability on $SU(2)^I$. The formalization constructs, without choosing an enumeration of I , the measurable equivalence

$$\begin{aligned} \Phi_I : SU(2) \times SU(2)^I &\rightarrow SU(2) \times SU(2)^I, \\ \Phi_I(x, y) &= (x, (x^{-1}y_i)_{i \in I}). \end{aligned}$$

with inverse $(u, v) \mapsto (u, (uv_i)_i)$. The product measure is Mathlib's `Measure.pi`. Using finite-product Haar invariance and the skew-product theorem, Lean proves

$$(\Phi_I)_*(dg \otimes d\mu_I) = dg \otimes d\mu_I \tag{5}$$

for every finite I . It also proves that the diagonal action $(x, y_i) \mapsto (ux, uy_i)$ preserves the same measure.

The result is consumed at the integral level. If $F : SU(2) \times SU(2)^I \rightarrow \mathbb{C}$ is invariant under that diagonal action, then

$$\int F(x, y) dg(x) d\mu_I(y) = \int F(1, y) d\mu_I(y). \tag{6}$$

More generally, the source contains an explicit factorization theorem for any density whose pullback through Φ_I^{-1} loses its first coordinate. The principal uniform endpoints are:

```
su2FiniteStarDiagonalLeft_measurePreserving
su2FiniteStarGaugeFix_measurePreserving
su2FiniteStar_integral_eq_gaugeFixed
su2FiniteStar_integral_eq_identitySlice
```

Equations (5)–(6) remove the fixed-valence limitation of the gauge step. Their composition requires a separate combinatorial theorem selecting and ordering a spanning tree. The next subsection closes that global coordinate theorem at the vertex level; facial holonomies remain a further edge-model statement.

6.2 Global rooted-tree vertex coordinates

The development now performs the global composition at the level of vertex coordinates. A value of `SU2RootedTreeOrder n` records a rooted tree on $n + 1$ vertices by adjoining one leaf at a time, together with the already present parent of each new leaf. Its parent–child list is embedded in the final index type. For every finite connected simple graph, Lean constructs an equivalence

$$e : \text{Fin}(n + 1) \simeq V$$

and such an order for which every recorded parent–child pair is an actual graph edge. The proof removes a vertex whose complement remains connected, recurses on the induced graph, and reattaches the vertex across a witnessed edge.

For vertex variables $x_0, \dots, x_n \in SU(2)$, the associated single global map retains the root and replaces each later variable by its tree increment:

$$\Psi_T(x)_0 = x_0, \quad \Psi_T(x)_k = x_{p(k)}^{-1} x_k \quad (k > 0). \tag{7}$$

It is implemented recursively by splitting the last coordinate, applying the triangular Haar shear, transforming the older coordinates, and reassembling the finite product. The inverse is part of the measurable equivalence, not an existence assertion. Induction and finite-product Haar invariance prove

$$(\Psi_T)_* \left(\bigotimes_{v \in V} dg_v \right) = \bigotimes_{v \in V} dg_v. \quad (8)$$

For a `SU2FiniteDiskCellulation`, primal adjacency is derived from an actual oriented half-edge and made into a simple graph. Under the explicit hypothesis that this primal graph is connected, Lean chooses the certified tree internally, transports (7) through the vertex enumeration, and returns a global measure-preserving equivalence on `C.Vertex → SU2`. The principal endpoints are:

```
SU2RootedTreeOrder.exists_validForGraph_of_connected
SU2RootedTreeOrder.coordinateEquiv_measurePreserving
SU2FiniteDiskCellulation.exists_rootedSpanningTree_of_primal_connected
SU2FiniteDiskCellulation.RootedSpanningTree.vertexCoordinateEquiv_measurePreserving
SU2FiniteDiskCellulation.exists_rootedTreeVertexCoordinateEquiv_measurePreserving
```

This closes global composition and product-Haar preservation for the vertex gauge coordinates. It does *not* yet prove the stronger edge-space decomposition

$$SU(2)^E \simeq SU(2)^{V \setminus \{r\}} \times SU(2)^{E \setminus T},$$

nor the simultaneous transformation formula for every facial holonomy. Those are the remaining ingredients for equality between the unreduced edge model and the schedule-independent amplitude for every cellulation.

7 Independent finite cellulations and elimination

The analytic evaluator used by the development is a labelled binary schedule. If a schedule node has left and right subproblems L, R , its amplitude is defined recursively by

$$A_C(a, b) = \int_{SU(2)} A_L(a, x) A_R(x^{-1}, b) dx.$$

The earlier tree module proves by structural induction that

$$A_C(a, b) = K_{T(C)}(ab).$$

At each node the induction hypotheses expose two heat kernels and the proof invokes the concrete two-face theorem.

The new point is that the geometric input is no longer this evaluator syntax. The new independent cellulation structure stores finite types of vertices, edges, half-edges, and bounded faces; a fixed-point-free involution pairs the two half-edges of every edge; source and target incidence is linked to a cyclic face-successor permutation; **none** denotes the exterior face; and the disk Euler relation $V - E + F = 1$ is part of the certificate. The dual face graph is derived from the two sides of internal edges.

A labelled elimination schedule is valid when its leaves list every bounded face exactly once and each binary merge is witnessed by a genuine adjacency in the derived dual graph. Lean proves the following existence theorem for an arbitrary finite connected dual graph. If the graph has one vertex, the schedule is a leaf. Otherwise, a vertex whose deletion leaves a connected induced subgraph is removed, the proof recurses on the smaller finite type, and the vertex is reattached across an actual adjacent edge. Thus no acyclicity assumption is used.

For any valid schedule S , permutation of the face list gives $T(S) = \sum_f t_f$. The tree reduction then proves

$$A_{C,S}(a,b) = K_{\sum_f t_f}(ab). \quad (9)$$

Consequently, for all valid S_1, S_2 ,

$$A_{C,S_1}(a,b) = A_{C,S_2}(a,b).$$

The public structure `SU2ConnectedDiskCellulation` stores only the cellulation and connect-
edness of its dual. A schedule is chosen internally from the existence theorem; the preceding
equality proves that every other valid choice is observationally identical. Its principal decla-
rations are:

```
SU2FaceEliminationSchedule.exists_schedule_of_connected_graph
SU2FaceEliminationSchedule.exists_valid_schedule_of_dual_connected
SU2FaceEliminationSchedule.amplitude_eq_of_valid_schedules
SU2ConnectedDiskCellulation.amplitude_eq_heatKernel
SU2ConnectedDiskCellulation.amplitude_eq_any_valid_schedule
```

For each representation label n , the formalization defines a plane-loop theory whose loops
are connected independently specified cellulations, whose area is $\sum_f t_f$, whose string ten-
sion is $n(n+2)/4$, and whose Wilson functional integrates the schedule-independent ampli-
tude. Equation (9) followed by coefficient extraction gives a nontrivial concrete instance
`su2ConnectedDiskExactAreaLawPackage`. The generic public interface is then consumed by
`su2ConnectedDisk_simpleLoop_areaLaw_exact`; no field of that package remains assumed.
The trivial-label case also proves the normalization

$$\int_{SU(2)} A_C(1,g) dg = 1.$$

For a general dual graph with cycles, this theorem concerns the explicitly defined post-
gauge-fixed schedule amplitude. The preceding section proves equality with the unreduced
product-Haar integral for the concrete three-cycle, but equality over every original edge of an
arbitrary cellulation is not yet asserted.

8 Exact simple-loop coefficient

The same repository proves, for every label n and positive area t , that the normalized Wilson
character $W_n = \chi_n/(n+1)$ extracts the corresponding heat coefficient:

$$\int_{SU(2)} W_n(g) K_t(g) dg = e^{-tc_n} = \exp\left[-t \frac{n(n+2)}{4}\right].$$

The proof first establishes the identity for partial character sums and then passes to the full
kernel by dominated convergence. Its final consumer is:

```
su2_exact_simpleLoop_areaLaw
```

Together with the iterated Migdal reduction, this is the exact two-dimensional route for
connected finite combinatorial disks after the certified elimination step: the choice of schedule
does not change the density, and the effective face has the Casimir coefficient prescribed by
its total area.

9 Audit and reproducibility

The checked source is built with Lean 4 and Mathlib. The public root module imports the concrete orbit, character-convolution, heat-semigroup, and exact area-law modules. A repository-wide placeholder audit finds no occurrence of `sorry`, `admit`, or a project-local axiom declaration in the Lean development. Printing axioms for the forty-three audited public endpoints returns only the standard kernel dependencies

`propext,      Classical.choice,      Quot.sound.`

The toolchain and Mathlib revision are pinned in the repository. GitHub Actions builds the public root target from a clean checkout and separately rejects placeholders.

Principal new declarations.

```
su2FirstRailMoment_recurrence
su2FirstRailMoment_eq
su2FirstRailMassMeasure_eq_uniform
intervalIntegral_chebyshevU_product_formula
integral_su2FirstEntryMass_eq_unitInterval
integral_su2Haar_orbit_character_general
integral_su2CharacterReal_mul_translate
su2CharacterChebyshev_convolution
su2HeatKernelPartial_convolution
su2HeatKernel_convolution
su2Migdal_twoFace_merge
su2Migdal_subdivision_invariant
su2_exact_simpleLoop_areaLaw
SU2PlanarFaceTree.internalEdgeCount_eq_faceCount_sub_one
su2PlanarCellulationAmplitude_eq_heatKernel
su2PlanarCellulationAmplitude_eq_effectiveFace
su2TreePlanar_wilsonExpectation_eq_casimir
su2TreePlanarExactAreaLawPackage
su2TreePlanar_simpleLoop_areaLaw_exact
integral_su2PlanarCellulationAmplitude_eq_one
SU2FaceEliminationSchedule.exists_schedule_of_connected_graph
SU2FaceEliminationSchedule.exists_valid_schedule_of_dual_connected
SU2FaceEliminationSchedule.amplitude_eq_of_valid_schedules
SU2ConnectedDiskCellulation.amplitude_eq_heatKernel
SU2ConnectedDiskCellulation.amplitude_eq_any_valid_schedule
su2ConnectedDisk_wilsonExpectation_eq_casimir
su2ConnectedDiskExactAreaLawPackage
su2ConnectedDisk_simpleLoop_areaLaw_exact
integral_su2ConnectedDiskAmplitude_eq_one
su2ThreeSpokeGaugeFix_measurePreserving
su2ThreeSpoke_density_gaugeFactorization
su2ThreeSpoke_gaugeFixedIntegral_eq_heatKernel
su2ThreeSpoke_unreducedIntegral_eq_gaugeFixedIntegral
su2ThreeSpoke_unreducedIntegral_eq_heatKernel
su2FiniteStarDiagonalLeft_measurePreserving
su2FiniteStarGaugeFix_measurePreserving
su2FiniteStar_integral_eq_gaugeFixed
su2FiniteStar_integral_eq_identitySlice
SU2RootedTreeOrder.exists_validForGraph_of_connected
SU2RootedTreeOrder.coordinateEquiv_measurePreserving
SU2FiniteDiskCellulation.exists_rootedSpanningTree_of_primal_connected
SU2FiniteDiskCellulation.RootedSpanningTree.vertexCoordinateEquiv_measurePreserving
SU2FiniteDiskCellulation.exists_rootedTreeVertexCoordinateEquiv_measurePreserving
```

10 Scope and next frontier

The present artifact closes the compact-group analytic chain, introduces an independent finite oriented disk-cellulation object, constructs elimination schedules for every connected dual graph, and proves independence of the schedule for the reduced amplitude. Dual cycles are allowed at this level, and the three-face cycle is additionally linked to its original product-Haar edge integral by a formal lattice gauge-fixing change of variables. The local measure-preserving change and identity-slice theorem hold for arbitrary finite valence. The global vertex-coordinate transformation is now constructed over a certified primal spanning tree and preserves the full vertex product Haar.

It does not yet lift this transformation to all original edge variables or identify all transformed face holonomies. It also does not identify the abstract combinatorial maps with planar isotopy classes, treat nonsimple or intersecting loops, construct a continuum measure, or evaluate higher-genus partition functions. The next precise frontier is therefore a uniform spanning-tree gauge-fixing theorem from the unreduced edge model of an arbitrary connected disk to the schedule-independent face model.

References

- [1] A. A. Migdal, *Recursion equations in gauge field theories*, Sov. Phys. JETP **42** (1975), 413–418.
- [2] E. Witten, *On quantum gauge theories in two dimensions*, Commun. Math. Phys. **141** (1991), 153–209.
- [3] B. K. Driver, *Yang–Mills theory and the heat equation on compact Lie groups*, J. Funct. Anal. **110** (1992), 1–34.
- [4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, Lecture Notes in Computer Science 12699 (2021), 625–635.
- [5] The Mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.